CSE422: Artificial intelligence

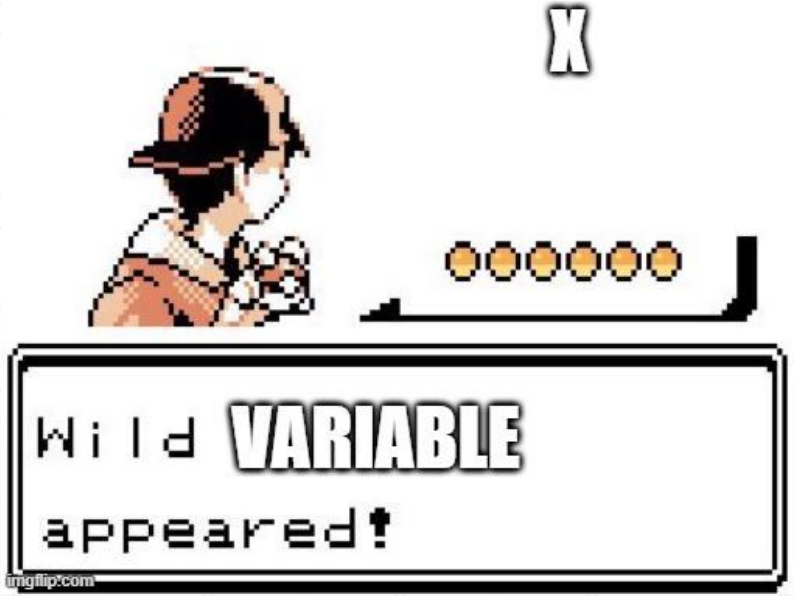
Probability

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You PROBABLY Know These

- Probability of an Event is defined by how LIKELY the event is to occur in an experiment
- If we repeat an experiment many times and record how often an event E happens, then probability of event E : P(E) = Number of times E occurs / Total number of trials
- In a coin toss, probability of heads occurring: P(head) = 1 / 2 = 0.5
- In a fair dice rolling (where all faces of the dice are equal)
    - Probability of getting 4: P(4) = 1 / 6
    - Probability of NOT getting 4: P(not 4) = 1 - P(4) = 1 - 1/6 = 5/6
    - Probability of getting an even number = P({2, 4, 6}) = P(2) + P(4) + P(6) = 3/6 = 0.5

Random Variables



Random Variables

- A **Random Variable (RV)** is an aspect of the environment for which we have some uncertainty
- A Random variable can have ONE of a set of values from its **domain**, each with an associated probability
    - Example: Weather is a RV with domain = {sunny, rainy, cloudy, snowy}
- Domain values must be exhaustive and mutually exclusive
- Random variable can be **DISCRETE** or **CONTINUOUS**
    - Discrete RV can take a discrete value from a finite range. Example: weather, mark you will get in the MID
    - Continuous RV can take ANY value from an infinite / continuous range. Example: Temperature of a day
- Let X be a RV defined as the SUM of two fair dice, what are the domain values?

Probability Distribution

- A probability distribution assigns a probability to each value of a Random Variable
- Let X be a random variable that represents YOUR CURRENT MODE, with the domain being {bored, sleepy, annoyed, excited, anxious about mid marks}
  - P(X = bored) = 0.25
  - P(X = sleepy) = 0.20
  - P(X = annoyed) = 0.10
  - P(X = excited) = 0.05
  - P(X = anxious about mid marks) = 0.4
- We write this as: P(X) = <0.25, 0.20, 0.10, 0.05, 0.4> [called Vector Notation]
- The sum of all probabilities in a distribution must be 1 : TRUE or FALSE?

Some Notations

- Suppose **weather** is a discrete RV with the domain {sunny, rainy, cloudy, snowy}
  - *"Weather = sunny"* is written as *"sunny"*
  - *"P(Weather = sunny) = 0.6"* is written as *"P(sunny) = 0.6"*
- Suppose cavity is a discrete RV with the domain {TRUE, FALSE}
  - *"Cavity = TRUE"* is written as *"cavity"*
  - *"Cavity = False"* is written as *"¬cavity"*
  - *"P(Cavity = False) = 0.6"* is written as *"P(¬cavity) = 0.6"*
- Probability distribution is often presented in a table
  - P(Weather) = <0.6, 0.15, 0.24, 0.01> is the same as:

Weather	Probability
sunny	0.60
rainy	0.15
cloudy	0.24
snowy	0.01

Joint Probability Distribution

- Probability distribution over a SET of Random variables (i.e. two or more random variables are being considered)
- Let's consider two boolean RVs: **Toothache** and **Cavity**

	toothache	¬toothache
cavity	0.22	0.15
¬cavity	0.08	0.55

- Sum of ALL entries in the joint probability distribution table must be 1
- Every question about a domain can be answered by looking at the table
- Quiz: P(cavity, toothache) = ? P(¬cavity, toothache) = ?
- Quiz: If I have cavity, probability of toothache is 0.22 -> TRUE or FALSE?

Marginal Probability Distribution

	toothache	¬toothache
cavity	0.22	0.15
¬cavity	0.08	0.55

- Probability of a proposition is the sum of probabilities of all events where that proposition holds TRUE
- P(cavity) = sum of all events where (Cavity=cavity) holds = 0.22 + 0.15 = 0.37
- P(¬toothache) = sum of all events where (Toothache=¬toothache) holds = 0.15 + 0.55 = 0.70
- Quiz: P(¬cavity) = ? P(toothache) = ?

AND OR

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cap B) = P(A) \times P(B)$ [ONLY IF A and B are independent, else not]
- $P(A \cup B)$ is also written as $P(A \vee B)$
- $P(A \cap B)$ is also written as $P(A \wedge B)$

Quiz on Probability Distribution

Let's consider a Joint probability distribution with THREE boolean RV: H, B, S

Quiz on Probability Distribution

Let's consider a Joint probability distribution with THREE boolean RV: H, B, S

H	*B*	*S*	Prob.
+h	+b	+s	0.44
+h	+b	-s	0.15
+h	-b	+s	0.14
+h	-b	-s	0.07
-h	+b	+s	0.10
-h	+b	-s	0.04
-h	-b	+s	0.05
-h	-b	-s	0.01

Quiz on Probability Distribution

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H	*B*	*S*	Prob.
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+h	-b	+s	0.14
+h	-b	-s	0.07
-h	+b	+s	0.10
-h	+b	-s	0.04
-h	-b	+s	0.05
-h	-b	-s	0.01

- P(+h, -b, +s) = 0.14
- P(+b) = 0.44 + 0.15 + 0.10 + 0.04 = 0.73
- P((+h, -b) OR +s)

  = P(+h, -b) + P(+s) - P(+h, -b, +s)

  = (0.14 + 0.07) + (0.44 + 0.14 + 0.10 + 0.05) - 0.14

  = 0.80

Conditional Probability

- Conditional probability is the probability of an event A occurring **given** that another event B has already occurred
- Written as P(a | b) [read as Probability of a given b] and defined by:
  - $P(a \mid b) = P(a \cap b) / P(b)$
  - Also written as $P(a \mid b) = P(a, b) / P(b)$
- Here b is **evidence** (we know its probability) and a is **query** (we WANT to know its probability)

	toothache	¬toothache
cavity	0.22	0.15
¬cavity	0.08	0.55

Probability of toothache IF i have cavity

= P(toothache | cavity)

= P(toothache, cavity) / P(cavity)

= 0.22 / (0.22 + 0.15)

= 22 / 37 = 0.595

Conditional Probability

	Male	Female
GoT	0.16	0.24
BB	0.2	0.05
TBBT	0.1	0.25

- P(GoT | Male) = ?
- P(Male | GoT) = ?
- P(TBBT | Female) = ?

Conditional Probability

	Male	Female
GoT	0.16	0.24
BB	0.2	0.05
TBBT	0.1	0.25

- P(GoT | Male) = P(GoT, Male) / P(Male) = 0.16 / 0.46 = 0.3478
- P(Male | GoT) = P(Male, GoT) / P(GoT) = 0.16 / 0.4 = 0.4
- P(TBBT | Female) = ? Do it yourself

Conditional Probability

	Right handed	Left handed
Male	0.41	0.08
Female	0.45	0.06

- From the joint probability distribution given above, find the probability that a randomly selected person is:
  - Male given right handed
  - Right handed given male
  - Female given left handed
  - Are the events “being female” and “being left handed” INDEPENDENT?

Conditional Probability

	Right handed	Left handed
Male	0.41	0.08
Female	0.45	0.06

- From the joint probability distribution given above, find the probability that a randomly selected person is:
  - Male given right handed [0.477]
  - Right handed given male [0.837]
  - Female given left handed [0.429]
  - Are the events “being female” and “being left handed” INDEPENDENT? [No]

Marginal Probability

	toothache		¬toothache	
	shows	¬shows	shows	¬shows
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

- Consider three Boolean RV Toothache, Cavity, and ShowsOnXRay
- P(cavity) = ?
- P(toothache) = ?
- P(cavity $\vee$ toothache) = ?

Marginal Probability via Conditioning

	toothache		¬toothache	
	shows	¬shows	shows	¬shows
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

- If A and B are two boolean RVs,
  - P(a) = P(a ∧ b) + P (a ∧ ¬b) [easier to find if Joint distribution table given]
  - In other words, P(a) = P(a | b) × P(b) + P(a | ¬b) × P(¬b) [if table not given]
- P(cavity) = P(cavity ∧ shows ∧ toothache) + P(cavity ∧ ¬shows ∧ toothache) + P(cavity ∧ shows ∧ ¬toothache) + P(cavity ∧ ¬shows ∧ ¬toothache)

Marginal Probability via Conditioning

Airport security is evaluating a new scanner by comparing its alerts against reality. For every passenger, **three** things are recorded: whether they are actually a **Smuggler**, whether they appeared **Nervous** on the walkthrough camera, and whether the scanner **Flagged** their bag. Here's what the records show:

- Among smugglers who appeared nervous: 5% were flagged, and 1% were not flagged.
- Among smugglers who appeared calm (i.e. did not appear nervous): 1% were flagged, and 3% were not flagged.
- Among innocent travelers (i.e. not smuggler) who appeared nervous: 8% were flagged, and 12% were not flagged.
- Among innocent travelers who appeared calm: 10% were flagged, and 60% were not flagged.

(a) What is the marginal probability that a randomly selected passenger is a **Smuggler but appeared calm (did not appear Nervous)?**

(b) What is the marginal probability that a randomly selected passenger is **actually innocent (not a smuggler)?**

Marginal Probability via Conditioning

Solution:
Let the random variables be S (Smuggler), N (Nervous) and F (Flagged). Each is a boolean variable: S = {s (smuggler), ¬s (innocent)}; N = {n (nervous), ¬n (calm)}; F = {f (flagged), ¬f (was not flagged)}

- Among smugglers who appeared nervous: 5% were flagged, and 1% were not flagged.
  - **P(s, n, f) = 0.05, P(s, n, ¬f) = 0.01**
- Among smugglers who appeared calm (i.e. did not appear nervous): 1% were flagged, and 3% were not flagged.
  - **P(s, ¬n, f) = 0.01, P(s, ¬n, ¬f) = 0.03**
- Among innocent travelers (i.e. not smuggler) who appeared nervous: 8% were flagged, and 12% were not flagged.
  - Find it yourself
- Among innocent travelers who appeared calm: 10% were flagged, and 60% were not flagged.
  - Find it yourself

(a) What is the marginal probability that a randomly selected passenger is a **Smuggler but appeared calm (did not appear Nervous)?** **-> P(s, ¬n) = P(s, ¬n, f) + P(s, ¬n, ¬f) = 0.01 + 0.03 = 0.04**

(b) What is the marginal probability that a randomly selected passenger is **actually innocent (not a smuggler)?** **-> P(¬s) = P(¬s, n, f) + P(¬s, n, ¬f) + P(¬s, ¬n, f) + P(¬s, ¬n, ¬f) = ...**

Marginal Probability via Conditioning

An insurance company audits claims by tracking **three** things for every filed claim: whether it is actually **Fraudulent**, whether the claimant used a **Rental Car** at the time of the incident, and whether the claim was **flagged** by the company's AI Auditor.

Here's what the records show:

- Among fraudulent claims involving a rental car: 6% were flagged, and 2% were not flagged.
- Among fraudulent claims with no rental car: 3% were flagged, and 1% were not flagged.
- Among honest claims involving a rental car: 9% were flagged, and 15% were not flagged.
- Among honest claims with no rental car: 5% were flagged, and 59% were not flagged.

(a) What is the marginal probability that **a claim is Fraudulent but was not flagged by the company's AI detector? [Ans: 0.03]**

(b) What is the marginal probability that a randomly selected claim is **Fraudulent**? **[Ans: 0.12]**

BAYES' Rule

- From the formula of conditional probability P(x | y) = P(x, y) / P(y), we can infer: P(x, y) = P (x | y) × P(y) [called the product rule]
- Similarly, P(x, y) = P (y | x) × P(x)
- Dividing we get:

$$P(x|y) = \frac{P(y|x)}{P(y)} P(x)$$

- In other words:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

- Why is this SO important?
  - It is hard to know the cause, but it is easy to see the effect
  - Using this formula, we can infer the cause by analyzing the effect

BAYES' Rule

Example 1.14 A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.01, the test result will imply he has the disease.) If 0.5 percent of the population actually has the disease, what is the probability a person has the disease given that his test result is positive?

BAYES' Rule

Example 1.14 A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.01, the test result will imply he has the disease.) If 0.5 percent of the population actually has the disease, what is the probability a person has the disease given that his test result is positive?

Solution: Let D be the event that the tested person has the disease, and E the event that his test result is positive. The desired probability $P(D|E)$ is obtained by

$$P(D|E) = \frac{P(DE)}{P(E)} = \frac{P(E|D)P(D)}{P(E|D)P(D) + P(E|D^c)P(D^c)}$$
$$= \frac{(0.95)(0.005)}{(0.95)(0.005) + (0.01)(0.995)}$$
$$= \frac{95}{294} \approx 0.323$$

Note: If A and B are boolean RV: $P(a \mid b) + P(\neg a \mid b) = 1$

$P(a, b) + P(a, \neg b) = P(a)$

BAYES' Rule

A person is brought in front of a jury. The jury finds the defendant guilty in 98% of the cases in which they indeed committed the crime; and it finds the defendant not guilty 97% of the cases when the defendant did not commit the crime. Only 0.8% of the population has committed a crime.

If a person is found guilty by the jury, what is the probability that person has truly committed the crime?

Answer: 49 / 235 = 0.2085

BAYES' Rule

Wilson's disease is a very rare disease, only about 0.005% people are affected with it (this is an imaginary value, actual rate is even lower). Unfortunately, during a routine medical checkup, your uncle has been diagnosed with it. The best hospital in Bangladesh is 97% effective in diagnosing this disease (if a person has it, the hospital detects it correctly). In 2% cases the hospital falsely detects it in a person (the person does not actually have it). Now, you are no medical expert, but you DO know probabilistic theory. Based on the information available to you, would you suggest your uncle starts treatment right away, or would you suggest your uncle performs a second test in Singapore to check if he actually has it? Explain with mathematical reasoning.

Ans: 2.42X10^-3

Chain Rule of Conditional Probability

- We know, $P(a, b) = P(a \mid b) \times P(b)$ [product rule]
- We can extend this using chain rule of multiplication
- $P(a, b \mid c) = P(a, b, c) / P(c)$
  - Hence, $P(a, b, c) = P(a, b \mid c) \times P(c)$
- $P(a \mid b, c) = P(a, b, c) / P(b, c)$
  - Hence, $P(a, b, c) = P(a \mid b, c) \times P(b, c) = P(a \mid b, c) \times P(b \mid c) \times P(c)$
- $P(a, b, c, d, e) = P(a \mid b, c, d, e) \times P(b, c, d, e)$

  $= P(a \mid b, c, d, e) \times P(b \mid c, d, e) \times P(c, d, e)$

  $= P(a \mid b, c, d, e) \times P(b \mid c, d, e) \times P(c \mid d, e) \times P(d, e)$

  $= P(a \mid b, c, d, e) \times P(b \mid c, d, e) \times P(c \mid d, e) \times P(d \mid e) \times P(e)$

Probabilistic Independence

- If two events A and B are independent, then the probability of A occurring does not depend on B
- So, P(A | B) = P(A)
- Hence, P(A ∧ B) = P(A | B) × P(B) = P(A) × P(B) [proved]
- This is also called **absolute independence**
- Consider the joint distribution of the RVs Toothache, Cavity, and ShowsOnXRay
    - Suppose we bring in another RV Weather = {sunny, cloudy, rainy, snowy}
    - Dental problems have no effect on weather
    - So, P(Weather = cloudy | toothache, cavity, shows) = P(Weather = cloudy)
    - P(Weather = cloudy, toothache, cavity, shows) = P(Weather = cloudy) × P(toothache, cavity, shows)
    - **Caution!** P(toothache, clavity) ≠ P(toothache) × P(cavity) [toothache and cavity are not independent]

Probabilistic Independence

	Male	Female
GoT	0.16	0.24
BB	0.2	0.05
TBBT	0.1	0.25

- Are Male viewers and GoT independent?
- P(Male ∧ GoT) = ? P(Male)= ? P(GoT) = ?
- Is P(Male ∧ GoT) = P(Male) × P(GoT)?

Conditional Independence

- A and B are **conditionally independent** given C, then the probability of A occurring **given** C **does not depend** on B, and vice versa
- In notation: P(A | B, C) = P(A | C)
  - Similarly, P(B | A, C) = P(B | C)
- Consider the OG boolean RVs Toothache, Cavity, Shows
- If I have cavity, probability of X-Ray showing it does not depend on whether I have toothache or not
  - P(+shows | +toothache, +cavity) = P(+shows | +cavity)
- Same independence holds if I do not have cavity
  - P(+shows | +toothache, –cavity) = P(shows | –cavity)

Conditional Independence

- Mathematically we can show that A and B are **conditionally independent** given C iff **P(A, B | C) = P(A | C) × P(B | C)**
- In that case we can decompose the joint distribution:
  - P(A ∧ B ∧ C) = P(A | C) P(B | C) P(C) [Note: P(A ∧ B ∧ C) is the same as P(A, B, C)]
  - Therefore, P(A ∧ B ∧ C) = [P(A ∧ C) × P(B ∧ C)] / P(C)
- Since the random variable *Shows* is conditionally independent of *Toothache* given *Cavity*
  - P(Shows ∧ Toothache | Cavity) = P(Shows | Cavity) × P(Toothache | Cavity)

Independence Check

- To prove that A and B are **absolutely independent**, we need to show that
  - **P(A, B) = P(A) × P(B)**

- To prove that A and B are **conditionally independent given C**, we need to show that:
  - **P(A, B | C) = P(A | C) × P(B | C)**

Conditional Independence

	toothache		¬toothache	
	shows	¬shows	shows	¬shows
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

- From this join distribution, find out whether the random variables Shows and Toothache are independent given Cavity
  - Is P(Shows ∧ Toothache | Cavity) = P(Shows | Cavity) × P(Toothache | Cavity)?
    - P(Shows ∧ Toothache | Cavity) = ?
    - P(Shows | Cavity) = ?
    - P(Toothache | Cavity) = ?

Conditional Independence

	toothache		¬toothache	
	shows	¬shows	shows	¬shows
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

- From this join distribution, find out whether the random variables Shows and Toothache are independent given Cavity
  - Is P(Shows ∧ Toothache | Cavity) = P(Shows | Cavity) × P(Toothache | Cavity)?
    - P(Shows ∧ Toothache | Cavity) = 0.54
    - P(Shows | Cavity) = 0.9
    - P(Toothache | Cavity) = 0.6

Exercise - 1

A cybersecurity company monitors suspicious login attempts using **two independent** detection systems: **Firewall Alert (F)** and **Behavioral AI Alert (B)**. Historical data shows that **8% of all login attempts are actually malicious.**

- If a login attempt is **malicious (M)**: P(F | M) = 0.8; P(B | M) = 0.70;
- If a login attempt is **legitimate (!M)**: P(F | !M) = 0.10; P(B | !M) = 0.15;

Assume that, given whether a login is malicious or legitimate, both alert systems operate independently. One login attempt is selected at random.

(a) Find the probability that the **Firewall Alert triggers** but the **Behavioral AI Alert does not trigger**, given that **the login attempt is malicious**. Ans: 0.24

(b) Find the probability that **a randomly selected login attempt triggers** both alerts. Ans: 0.0586

(c) Suppose a login attempt **triggered** the **Firewall Alert** but **did not trigger the Behavioral AI Alert**. What is the probability that the **login attempt was malicious**? Ans: 0.1971

Exercise - 2

	smart		¬smart	
	study	¬study	study	¬study
prepared	.432	.16	.084	.008
¬prepared	.048	.16	.036	.072

Queries:

- What is the prior probability of smart?
- What is the prior probability of study?
- What is the conditional probability of prepared, given study and smart?
- Save these answers for later! ☺

Exercise

	smart		¬smart	
	study	¬study	study	¬study
prepared	.432	.16	.084	.008
¬prepared	.048	.16	.036	.072

Queries:

- Is smart independent of study?
- Is prepared independent of study?

Exercise

	smart		¬smart	
	study	¬study	study	¬study
prepared	.432	.16	.084	.008
¬prepared	.048	.16	.036	.072

Queries:

- Is smart conditionally independent of prepared, given study?
- Is study conditionally independent of prepared, given smart?